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A multi-modal pre-training transformer for universal transfer learning in metal–organic frameworks

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Metal–organic frameworks (MOFs) are a class of crystalline porous materials that exhibit a vast chemical space owing to their tunable molecular building blocks with diverse topologies. An unlimited number of MOFs can, in principle, be synthesized. Machine learning approaches can help to explore this vast chemical space by identifying optimal candidates with desired properties from structure–property relationships. Here we introduce MOFTransformer, a multi-modal Transformer encoder pre-trained with 1 million hypothetical MOFs. This multi-modal model utilizes integrated atom-based graph and energy-grid embeddings to capture both local and global features of MOFs, respectively. By fine-tuning the pre-trained model with small datasets ranging from 5,000 to 20,000 MOFs, our model achieves state-of-the-art results for predicting across various properties including gas adsorption, diffusion, electronic properties, and even text-mined data. Beyond its universal transfer learning capabilities, MOFTransformer generates chemical insights by analyzing feature importance through attention scores within the self-attention layers. As such, this model can serve as a platform for other MOF researchers that seek to develop new machine learning models for their work.

Owing to properties such as large surface area[1], high chemical/thermal stability[2] and tunability[3], crystalline porous MOFs are used for various energy and environmental applications[4–7]. Given that MOFs are composed of thousands of tunable molecular building blocks (that is, metal nodes and organic linkers), an infinite number of MOFs can, in principle, be synthesized. To efficiently explore this vast MOF search space, it is important to identify the structure–property relationships for a given application. One can then focus on MOFs that contain specific structures for user-desired properties. To gain such information, high-throughput computational screening approaches have been used to conduct simulations on a large dataset of MOF structures and retroactively identify the structure–property relationship[8–11]. However, this can be cumbersome and, more importantly, one would need to conduct independent computational screenings for each of the applications, requiring vast computational resources.

An alternative approach to discover structure–property relationships is through machine learning, which has gained a lot of traction recently[12,13]. In particular, geometric descriptors of MOF structures (for example, void fraction and pore volume) have been used to accurately predict various gas adsorption properties[14–16]. A machine learning model has also been developed[17] using energy grid histograms as descriptors to predict gas uptake properties. For diffusion properties, a machine learning model has been developed[18] to predict N_2/O_2 selectivity and diffusivity using geometric, atom-type and chemical feature descriptors. For electronic properties, it has

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been demostrated[19] that a graph neural network can facilitate capturing the underlying chemical features leading to accurate predictions of the band gap values for the MOFs. Unfortunately, in all these previous studies, the developed machine learning model cannot be readily transferred from one application to another. As such, one would need to restart the training process and develop a new machine learning model from scratch for every different application.

To remedy this issue, one can utilize transfer learning, which incorporates knowledge from one machine learning application to another and, thereby, in principle, saving computational time for subsequent machine learning. Although transfer learning has been applied in a few cases for MOFs, these were limited to specific properties (for example, transfer knowledge from gas uptake to gas diffusivity or between different gas types)[16,20]. To enable transferability across a wide range of properties, it requires a universal transfer learning model that can be applied to all possible properties. To achieve this, machine learning models and descriptors should capture two disparate MOF features: (1) local features (for example, specific bonds and chemistry of the building blocks) and (2) global features (for example, geometric and topological descriptors). Although both local descriptors (for example crystal graph convolutional neural networks (CGCNNs)[19,21], chemical descriptors[18], revised autocorrelation functions (RACs)[22,23] and building-block embedding[11,24,25]) and the global features (for example, geometric features calculated by ZEO++ (ref. [26]) and histograms of energy-grids[16,17]) have been developed previously, to our knowledge, none of these works have effectively captured both the local and global features to achieve universal transfer learning.

When it comes to multi-modal learning that takes in multiple inputs, the Transformer architecture[27] (initially proposed for sequence data, such as language models) has emerged as the dominant modelling network. Given that Transformer consists of self-attention layers, which handle data sequences in parallel, it facilitates efficient training of neural networks with vast amounts of data. In 2019, Google introduced BERT, a pre-training Transformer encoder in the language model[28], and demonstrated remarkable performance in transfer learning. By fine-tuning the pre-trained BERT model, it obtained state-of-the-art performance results for many natural language process tasks such as question answering and named entity recognition. Moreover, for computer vision, various vision Transformer architectures have emerged as an alternative to convolution neural networks[29]. Recently, the pre-trained Transformer's transfer learning strategy has been expanded to multi-modal learning[30]. This pre-trained, multi-modal Transformer achieved state-of-the-art results in vision–language models such as image captioning and vision question answering[31–33]. Owing to its superior performance, the Transformer architectures have recently been adopted to predict various properties of MOFs[34,35].

Here, we introduce a multi-modal Transformer architecture, 'MOFTransformer', which captures both local and global MOF features. Our MOFTransformer was pre-trained with 1 million hypothetical MOFs (hMOFs). Fine-tuning the pre-trained MOFTransformer showcases excellent prediction capabilities across multiple different properties (for example, gas uptake, gas diffusivity, electronic properties of MOFs and text-mined data). Aside from the performance, this architecture allows chemists to capture insights from attention scores obtained from the attention layers of the MOFTransformer. As such, we believe that this model can serve as a bedrock architecture or model for future machine learning research by the MOF community.

Results

MOFTransformer

The overall schematics of our MOFTransformer are shown in Fig. 1a. To build towards universal transfer learning, we implement both pre-training and fine-tuning strategies. The objective of pre-training is to allow MOFTransformer to learn the essential characteristics of a MOF. This pre-trained model serves as a starting point for all subsequent applications. Fine-tuning refers to the process of training the pre-trained models for the specific application at hand (for example, gas adsorption uptake prediction). Figure 1b shows a schematic of the MOFTransformer architecture, which is based on a multi-layer, bidirectional Transformer encoder previously[27]. MOFTransformer is a multi-modal Transformer that takes two types of embedding as inputs, each representing the local and global features: (1) atom-based graph embedding and (2) energy-grid embedding.

Previously, a CGCNN was devised[21] that transforms atoms (nodes), bonds (edges) and their features (the distance between atoms) into a vector space. Although CGCNN consists of convolutional layers and pooling layers from the original paper, the atom-based graph embedding in MOFTransformer uses output vectors of CGCNN without the pooling layers. This allows our model to deal with the atom-wise features without losing information. It should be noted that many atoms in the unit cell of MOFs have the same embedding from CGCNN, given that CGCNN creates the embedding by taking atom types of nodes, distances and atom types of the neighbour nodes (Supplementary Fig. 1). We grouped topologically identical atoms and defined these sets as unique atoms (the details of the algorithm are explained in Supplementary Note 1). Removing the information from the overlapping atoms enables efficient training and prevents significant memory issues that frequently appear when training with long sequences of inputs.

For energy-grid embedding, the energy grids were calculated using a methane molecule probe that was selected for its facility in modelling. Universal Force Field[36] and TraPPE[37] were used to describe adsorbate–adsorbent van der Waals interactions in MOFs and the methane molecule, respectively. The 3D energy grids can be treated as 3D images, which means that the grid points and the energy values of the energy grids serve as pixels and one-channel colours, respectively. Similar to Vision Transformer[29], MOFTransformer takes 1D patches of the flattened 3D energy grids where (H, W, D) are the height, width and depth of energy grids and (P, P, P) is the patch resolution, and $N = HWD/P^3$ is the number of patches. Given that the energy grids were interpolated to 30 × 30 × 30 Å, H, W and D are 30 Å. The patch size P was set to 5 Å, so N is 216.

The MOFTransformer model is derived from the BERT-based model[28] ($L = 12$, $H = 768$, $A = 12$), where L is the number of blocks, H is the hidden size and A is the number of self-attention heads. Similar to BERT's class and separate tokens, the class token [CLS] and the separate token [SEP], which are learnable embedding layers, are located at the first position and between the two types of embedding, respectively (Fig. 1b). The [CLS] token is a head token of the Transformer blocks and predicts desired properties by adding a multi-layer perception (MLP) head which consists of a single pooling layer for the pre-training and fine-tuning tasks. Apart from these, a volume token [VOL], which is the normalized cell volume, is added at the final position of the input embedding because the interpolation of the energy grids leads to a loss of information regarding the volume of the original energy grids. Finally, position embedding and modal-type embedding, which are also learnable embedding layers, are added to the input embedding by the element-wise summation. The position embedding is a vector that encodes the position of the sequence, and the modal-type embedding encodes the two types of embedding to 0 and 1.

Understanding MOF descriptors

It is important to recognize how MOF descriptors (that is, local features and global features) influence the properties of MOFs. As shown in Fig. 2, H_2 uptake, H_2 diffusivity, and band gap were selected as case-study applications for MOFs that represent adsorption, diffusion, and electronic properties, respectively. Figure 2a–c shows the structure–property maps obtained from the molecular simulations for each of these applications. For H_2 uptake and diffusivity, the data were taken from our fine-tuning dataset (20,000 structures). The band gap values

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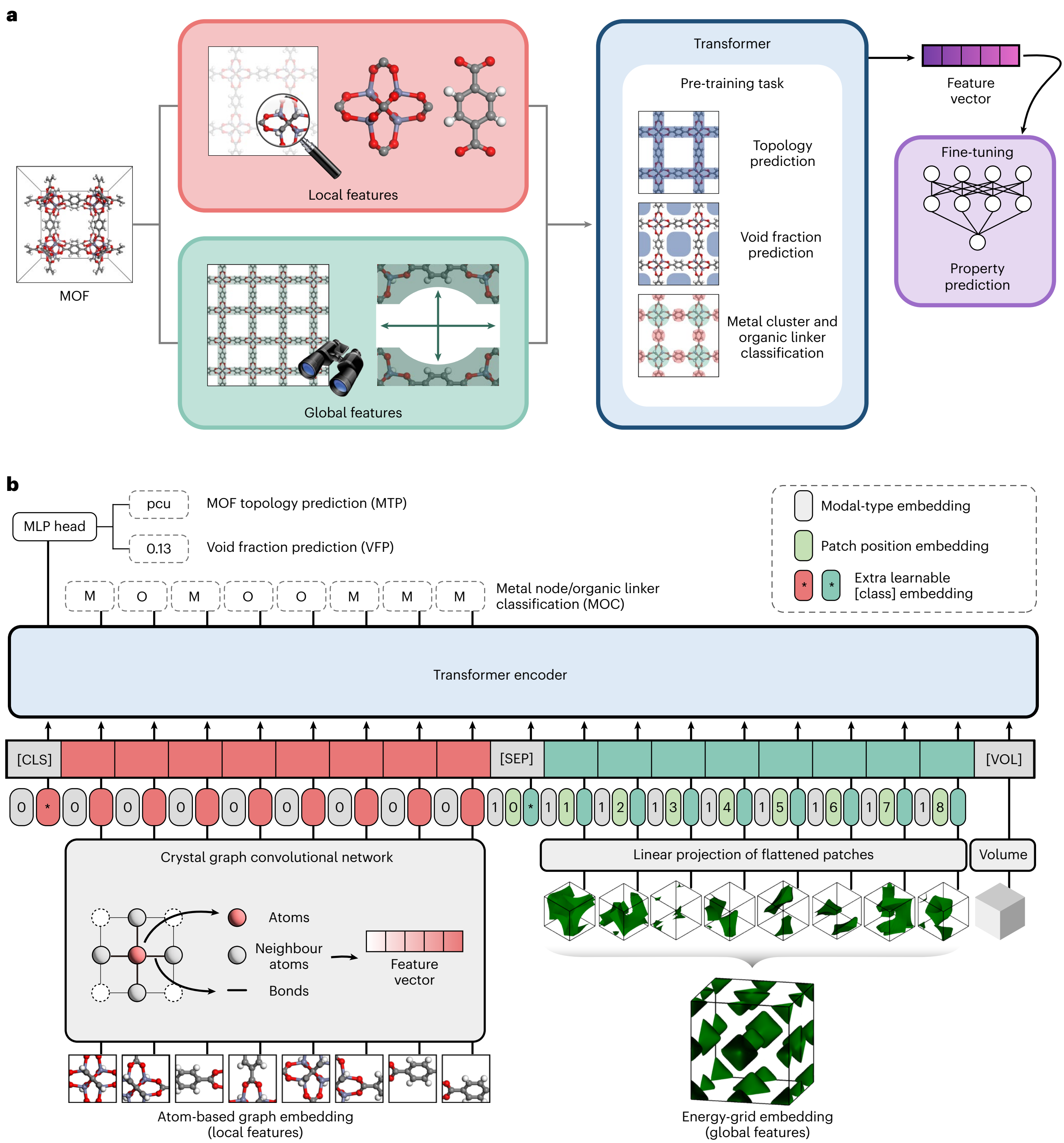

Fig. 1 | Overall schematics and architecture of MOFTransformer. a, Overall schematics of MOFTransformer. The model takes both local and global features as inputs. In the pre-training step, it undergoes three pre-training tasks. In the fine-tuning step, the model is trained to predict desired properties of MOFs using initial weights from the pre-trained model. **b**, Architecture of MOFTransformer. The input embedding takes atom-based graph embeddings and energy-grid embeddings that serve as local and global features, respectively. Binoculars image adapted with permission from iconfinder.com under a Creative Commons license CC BY 3.0.

are obtained from the QMOF database (version 13)[38] with the PBE functional that includes a total of 20,373 structures. From Fig. 2a,b, it can be seen that the H_2 uptake and diffusivity increase with accessible volume fraction and are strongly dependent on the MOF topology due to the correlation between topology and void fraction. Meanwhile, the band gap exhibits no correlation with accessible volume fraction and topology, which is reasonable given that electronic properties are more dependent on local chemical features as opposed to global geometric features.

Additionally, Fig. 2d–f shows the correlation between the MOF properties and the types of metal atoms. It can be seen that the dependence on metal atoms is lowest for H_2 uptake while highest for the band gap energy. Similar trends are observed for the organic linkers

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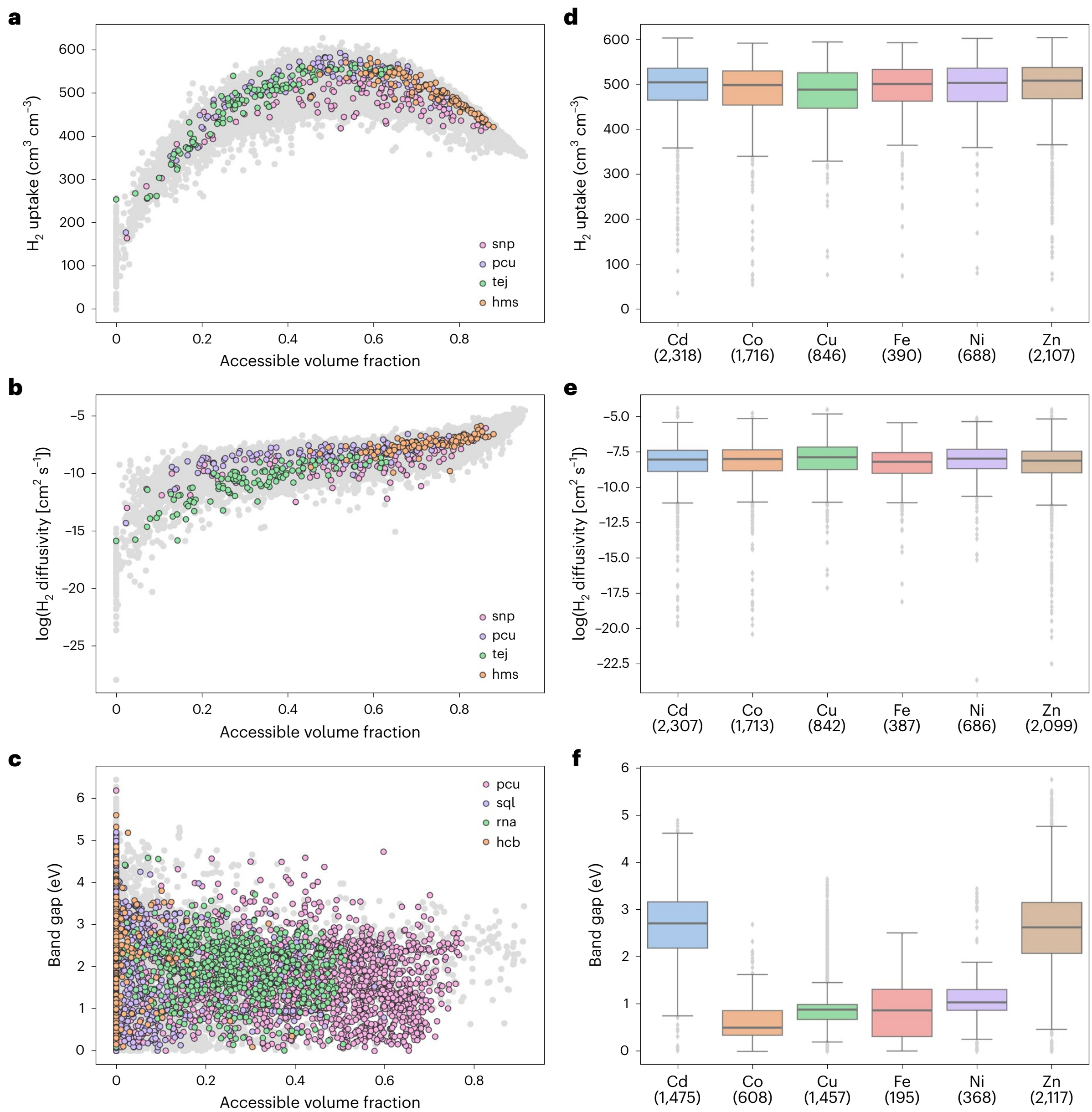

Fig. 2 | Relationship between global and local features and MOF properties. **a**–**c**, Scatter plots showing the relationship between accessible volume fraction and various properties (that is, gas uptake (**a**), diffusivity (**b**) and bandgap (**c**)). Grey dots represent the MOFs from each database, and colored dots represent MOFs with the top four topologies. **d**–**f**, Box plots of properties (adsorption (**d**), diffusion (**e**) and band gap (**f**)) for each metal type. The numbers in parentheses represent the number of data points. Boxplots show maximum and minimum values (whiskers), upper and lower quartiles (box boundaries) and median values (horizontal lines). The interquartile range, which is the spread between the upper and lower quartiles, covers 50% of the values.

(Supplementary Fig. 2). Along with the aforementioned geometric analysis, Fig. 2d–f confirms that adsorption and diffusion properties rely more on global features, while electronic properties rely more on local features. Apart from these, some properties, like O_2 diffusivity (which is more dependent on electronic effects than H_2 diffusivity) and CO_2 Henry coefficient, have more complex correlations between features and properties (Supplementary Fig. 3). This illustrates the importance of integrating both local and global features within the Transformer to enable universal transferability across different applications.

Pre-training results

The pre-training tasks play an essential role in determining the performance of the transfer learning. Three pre-training tasks were designed to capture the essential features of the MOFs: (1) MOF topology prediction (MTP), (2) void fraction prediction (VFP), and (3) metal cluster–organic linker classification (MOC). For the MTP task, the model was trained to predict the 1,079 topologies of MOFs by adding a classification head, which consists of a single dense layer to the [CLS] token. The list of topologies is summarized in Supplementary Table 1. For the VFP task, the model is trained to predict accessible void fraction calculated by ZEO++ (ref. [26]) by adding single dense layers to the [CLS] token. Finally, the MOC task was performed to enable the model to learn the features stemming separately from each metal node and organic linker. Binary classification (determining a given MOF atom as belonging to the metal or the organic linker) was conducted for the atom-wise features of atom-based embedding. The accuracies of MTP and MOC were 0.97 and 0.98, respectively, and the mean absolute error (MAE) of VFP was 0.01.

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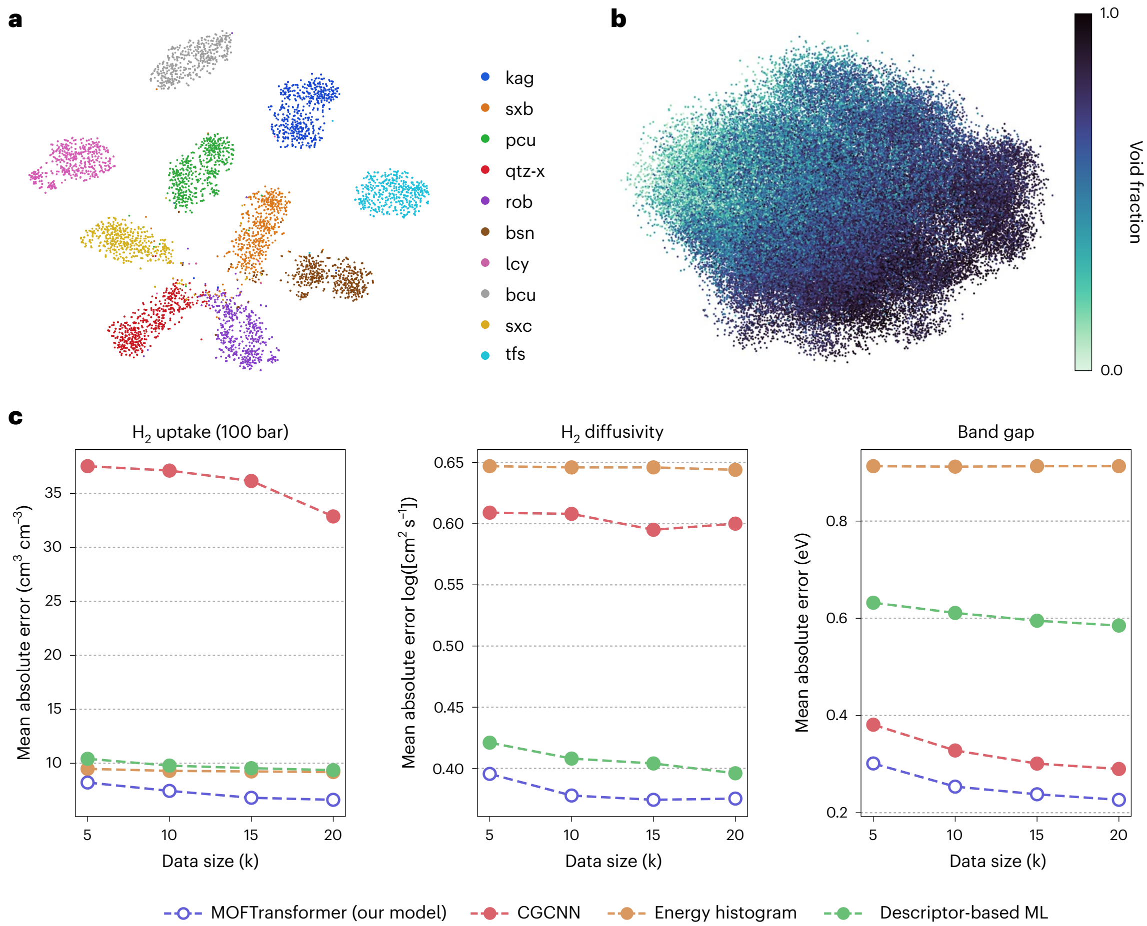

Fig. 3 | Results of pre-training and fine-tuning. a, For the top 10 frequently appearing topologies, the t-SNE plot embeds the class tokens of the pre-training model. **b**, The class tokens of the pre-trained model are embedded by PCA. Colour indicates void fraction. **c**, MAE results of the fine-tuning model and three baseline models with datasets of H_2 uptake, H_2 diffusivity and band gap according to dataset size. The baseline models are machine learning models that were respectively used to predict gas uptake, diffusivity and band gap values.

Next, we visualized the embedding vector of the pre-training model in 2D space, using t-distributed stochastic neighbor embedding (t-SNE) and principal components analysis (PCA) (Fig. 3). Figure 3a shows a t-SNE plot for the embedding vector of class tokens with the top 10 frequently appearing topologies in the dataset. Figure 3a shows that MOFs with different topologies are clustered together and segregated from other MOFs, indicating that proper learning has occurred. The same pattern of results was seen for all topologies (Supplementary Fig. 4). Furthermore, it is interesting to note that the PCA plots exhibit the distribution of the embedding vector that gradually increases according to the void fraction, as shown in Fig. 3b. This indicates that the embedding vectors are clustered with similar values of void fraction. These results demonstrate that the pre-training model is successfully trained to capture the critical features of the MOFs.

Fine-tuning results

Figure 3c shows the fine-tuning results for predicting H_2 uptake (100 bar), H_2 diffusivity and band gap, which were obtained from grand canonical Monte Carlo (GCMC), molecular dynamics (MD), and density functional theory (DFT) simulations, respectively. While 1 million hMOFs were used for the pre-training step, relatively smaller numbers (5,000–20,000) were used for training during the fine-tuning stages. The performance of the fine-tuning is compared with the three baseline models (energy histogram[17], descriptor-based machine learning model[18] and CGCNN[19,21]) as these have shown high performance in predicting gas uptake, diffusivity and band gap, respectively. From these comparisons, it can be seen that MOFTransformer outperforms all of these other models, demonstrating both its superior performance as well as transferable capabilities. It is worth noting that the Materials Graph Network (MEGNET)[39] outperforms the CGCNN in predicting the band gaps of MOFs[40]. MEGNET utilizes global state attributes such as system temperature as well as atomic and bond attributes as inputs. However, graph network models like CGCNN and MEGNET may have difficulty in effectively predicting properties that rely on global features such as gas uptake and diffusivity for MOFs. This is due to the larger crystal system of MOFs, which is characterized by a larger number of atoms and defined by metal clusters and organic linkers as nodes and edges, respectively. As a result, the MOFTransformer exhibits strong performance in universal transfer learning for MOFs compared to graph network models. The ablation studies of the fine-tuning to demonstrate the effect of the data size on the pre-training tasks are explained in the Supplementary Note 2.

To demonstrate further transferability across different applications, MOFTransformer was fine-tuned for various properties (Table 1). Performance comparison with previously reported machine learning models shows that MOFTransformer has similar or higher performance (that is, higher R^2 score or lower MAE) across all properties (Table 1). In particular, it is worth noting the robustness of our model across different gas types, even though the probe molecule used to generate energy grids was CH_4. The reason is that overall shape of energy grids is relatively insensitive to the type of probe molecule which has little effect on our model to learn global features from energy-grid embeddings.

Table 1 | Fine-tuning results with the publicly accessible databases of MOFs that include the properties calculated by GCMC, MD and even text-mining data

Property	MOFTransformer	Original paper	Number of data	Remarks	Ref.
N_2 uptake	R^2: 0.78	R^2: 0.71	5,286	CoREMOF	18
O_2 uptake	R^2: 0.83	R^2: 0.74	5,286	CoREMOF	18
N_2 diffusivity	R^2: 0.77	R^2: 0.76	5,286	CoREMOF	18
O_2 diffusivity	R^2: 0.78	R^2: 0.74	5,286	CoREMOF	18
CO_2 Henry coefficient	MAE: 0.30	MAE: 0.42	8,183	CoREMOF	22
Solvent removal stability classification	Accuracy: 0.76	Accuracy: 0.76	2,148	Text-mining data	40
Thermal stability regression	R^2: 0.44 (MAE: 45 °C)	R^2: 0.46 (MAE: 44 °C)	3,098	Text-mining data	40

The results of the machine learning models used in the paper on the databases are summarized to compare the performance.

In addition, MOFTransformer can accurately predict properties at ambient conditions, given that N_2, O_2 uptake and diffusivity were calculated at 1 bar and 298 K. Moreover, our model extends well to showcase lower MAE than the machine learning model using RACs[41] with geometric features as descriptors to predict solvent removal stability and thermal stability collected by text-mining. It is worth highlighting that our model showcases high performance in predicting the experimental data like text-minded data as well as the calculated properties. This result suggests that one can easily obtain high-performance structure–property relationships by using our pre-trained model and fine-tuning it without needing to develop a new model from scratch.

Discussion

Apart from universal transfer learning, feature importance and its interpretation can lead to a better understanding of the relationship between the MOF structures and their properties. Given that attention scores measure how much the model should pay attention to inputs when predicting desired properties, attention layers of the Transformer were assigned high attention scores to input features according to their importance. From the fine-tuning models that predict H_2 uptake, H_2 diffusivity, and band gap, feature importance analysis was implemented for IRMOF-1, which is one of the representative isoreticular MOFs. Figure 4 shows both the repeating building blocks models, which represent the metal cluster and organic linker, of IRMOF-1 (representing local features) and the 6 × 6 × 6 patches of energy grids (representing global features). The sizes of atoms in the repeating building block models are scaled according to the attention scores obtained by the atom-based embeddings. And the colours of the patches are proportional to the attention scores obtained from the energy-grid embeddings. As can be seen from Fig. 4, the atom-based embeddings are assigned with low attention scores (for example, visualized by small atom sizes) when predicting H_2 uptake and diffusivity. On the other hand, the energy-grid embeddings are assigned with high attention scores, which is in accordance with the fact that H_2 uptake and diffusivity rely more on the global features. Meanwhile, for the band gap prediction, there is a reversal in trend as the atom-based graph embeddings have higher attention scores compared to energy-grid embeddings as the band gap is more dependent on the local features. The additional feature importance analysis for other properties (for example, O_2 diffusivity and CO_2 Henry coefficient) were also conducted (Supplementary Fig. 8). Note that the feature importance analysis via attention scores is in line with previous findings and a chemist's intuition.

Beyond the case study of IRMOF-1, we implemented an in-depth analysis of feature importance for the atom-based graph and the energy-grid embeddings for band gap and H_2 uptake, respectively. Given that the band gap is defined by the difference between the conduction-band minimum (CBM) and the valance-band maximum (VBM), one might think that the atoms that exhibit strong peaks at the CBM and VBM play a critical role in determining its value. Interestingly, we identified that the atoms with peaks at the CBM and VBM strongly correlate with the atoms having high attention scores. Figure 5a shows the repeating building block models of IRMOF-1, 2, 3 and Ni-IRMOF-1 and their density of state (DOS) plots. IRMOF-2 and IRMOF-3 are variants of the IRMOF-1 structure with the BDC linker functionalized by −Br and $-NH_2$. For IRMOF-2 and IMROF-3, the atoms that are part of the organic linkers (that is, C, H, N, Br) have higher attention scores than those from the metal clusters (that is, Zn, O). Consistent with these results, the atoms of the organic linker have peaks at the CBM and VBM compared with those of the metal clusters. For the Ni-IRMOF-1 (which has Ni in place of Zn), the atoms that belong to the metal cluster have higher attention scores and stronger peaks at the CBM and VBM compared to the organic linkers. These tendencies are consistent with other examples that were randomly selected in the QMOF database (Supplementary Fig. 9). Apart from these, we confirmed that the feature importance analysis could capture the underestimation of the band gap calculated by the PBE functional (Supplementary Note 3). Hence, these results demonstrate that the fine-tuned model successfully learns the chemical features that are the more important when it comes to the band gap predictions.

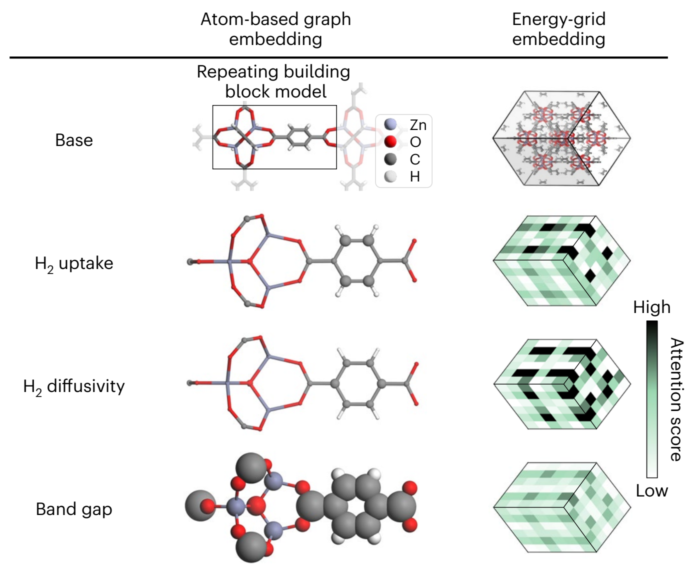

Fig. 4 | Schematics of attention scores in IRMOF-1. The schematics for attention score of atom-based embedding and energy-grid embedding in IRMOF-1. Left, repeating building block models in IRMOF-1 with atomic size proportional to attention score. Right, energy grids that represent attention scores by colour. The original form of the IRMOF-1 is shown in the 'base'.

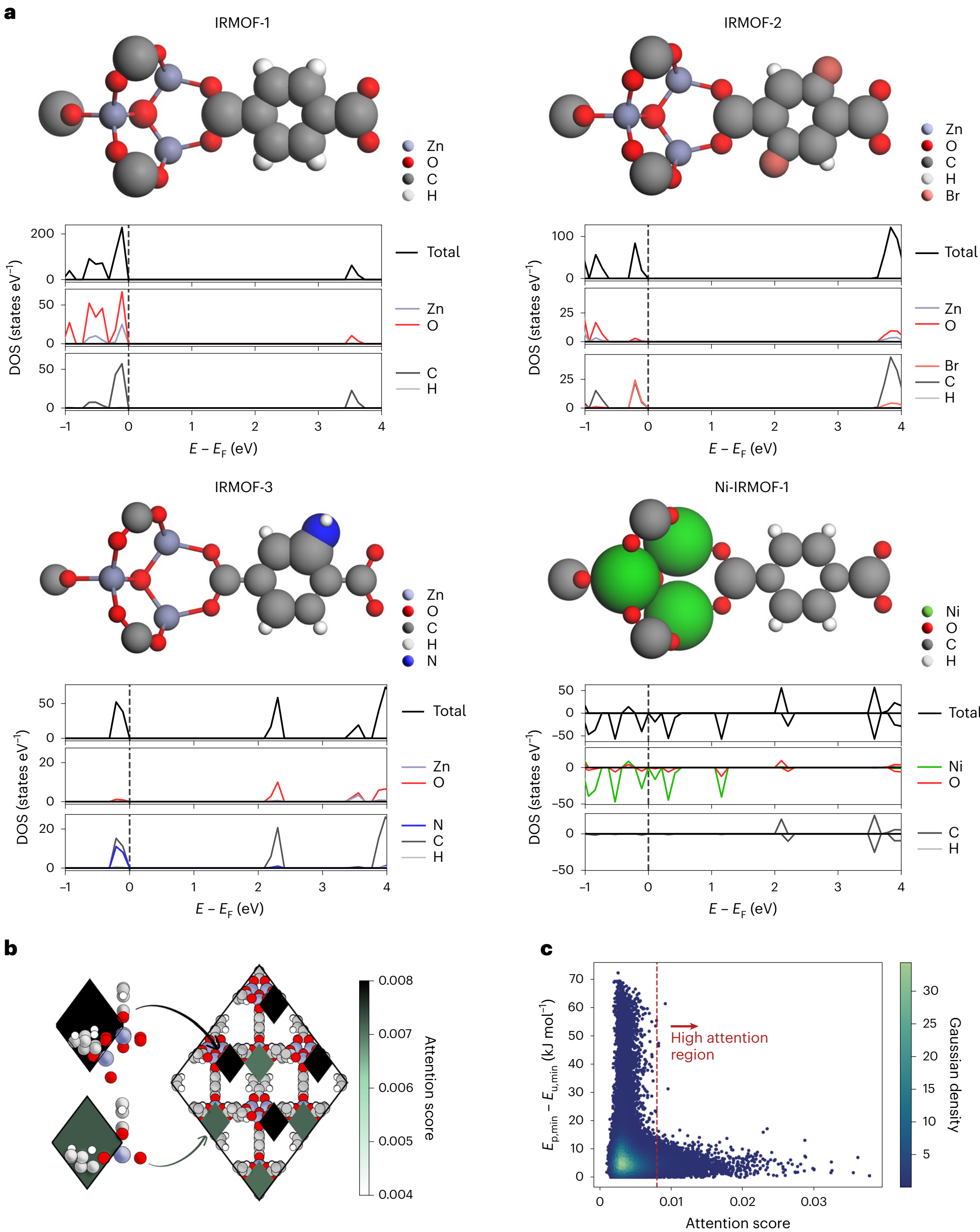

Fig. 5 | Feature importance analysis with attention scores. a, Schematics of attention score for atom-based embedding, and DOS plots for IRMOF-1, 2, 3 and Ni-IRMOF-1. The atomic sizes of repeating building block models are proportional to the attention score. *E*, energy; E_F, the Fermi level. Positive and negative DOS values indicate spin-up and spin-down channels, respectively. **b**, Schematic of high attention score patches of energy-grid embedding for IRMOF-1. **c**, Scatter plot for the difference in minimum energy between patch and unit cell according to energy-grid embedding. $E_{p,min}$, minimum energy of the patch; $E_{u,min}$, minimum energy of the unit cell. The red line ($x = 0.008$) distinguishes between high- and low-attention regions.

When it comes to the energy-grid embeddings, one could argue that the patches located near the metal atoms have an important role on determining the gas uptake[42]. Indeed, from the fine-tuned model to predict H_2 uptake, the eight highest attention scores from the 6 × 6 × 6 energy-grid patches of IRMOF-1 are located near the metal atoms as shown in Fig. 5b. The metal atoms can make stronger bonding with adsorbates than other atoms such as C, H and O, resulting in lower energy values for energy-grid patches near the metal atoms. Based on these observations, one can infer that the energy values of energy-grid patches can have an impact on determining attention score. Therefore, we plotted the relationship between the energy values of energy-grid points and the attention scores for each patch to further illustrate this relationship. The minimum energy values are normalized by their corresponding structure (or unit cell), which is represented on the *y*-axis of Fig. 5c. Figure 5c suggests that the energy-grid points with high attention scores tend to have relatively low energy values, as seen in the patches near the metal atoms. It is essential to highlight the fact that the scatter points within the high attention region (attention score >0.008) exhibit a lower difference of energy than 20 kJ mol^{-1}.

Conclusions

We have introduced a multi-modal, pre-trained Transformer to capture both local and global features of MOFs. The model facilitates capture of the chemistry of metal nodes and organic linkers from CGCNN and information on geometric and topological features, such as pore volume and topology, from energy grids. By fine-tuning the MOFTransformer model, our model outperforms other state-of-the-art machine learning models across various different properties, showing its universal transferability as well as superior performance. Moreover, the model can provide insights by analysing the feature importance from attention scores obtained from attention layers of the fine-tuned model. We believe that this model can be used as a bedrock model for other MOF researchers who wish to start their machine learning work and, as such, can help accelerate materials discovery and research within the field of porous materials.

Methods

Construction of hMOFs

The hMOFs used to train our MOFTransformer were constructed using PORMAKE[11], a Python library that can generate MOFs by combining building blocks with different topologies. These building blocks and the topologies were obtained from ToBaCCo[43], CoREMOF (with all of the solvents removed)[44] and RCSR database[45]. In total, 1 million and 20,000 hMOFs were generated for the pre-training and fine-tuning datasets, respectively, and the details of building hMOFs are explained in Supplementary Note 4. All of the generated structures were geometrically optimized using the LAMMPS[46] package with the UFF force field[36].

Computational details for molecular simulation

For the fine-tuning dataset, H_2 uptake and diffusivity (or diffusion coefficient) were selected to represent adsorption and diffusive properties. H_2 was selected to enable facile calculation while being different from the guest molecule (methane) used for the energy grid construction. The calculations were conducted using the RASPA package[47]. For the H_2 molecule, a united atom model was adopted. Also, the pseudo-Feynmna–Hibbs model was used to express the H_2 behaviour at low temperature, which leads to fitting the Lenard–Jones potentials to Feynman–Hibbs potential at $T = 77$ K (ref. [48,49]). Except for the H_2 molecules, the UFF force field was used with the Lorentz–Berthelot mixing rule and a cut-off distance of 12.8 Å.

For H_2 uptake calculation, the GCMC calculation was performed at 100 bar and 77 K for 10,000 production cycles with 5,000 cycles used for the initialization. Diffusivity (or diffusion coefficient) was calculated at infinite dilution at 77 K using the MD simulation. Given that the intermolecular interactions of the H_2 atoms are ignored for the infinite dilution simulation, it may sometimes lead to the initial configurations of the H_2 atoms captured within the small pores of MOFs. The initial configurations were obtained from the MC simulation without infinite dilution for 5,000 cycles to prevent this from happening. Then, the MD simulations were conducted by NVT ensemble with 1 fs time step[18,50]. The simulations were run for 3 million cycles, with 1,000 cycles used for the initialization and 10k cycles for equilibration. The guest molecules' mean-squared displacement (MSD) was computed every 10k cycles, and the diffusion coefficient was obtained using the slope of the MSD through Einstein's relation[51].

Pre-training and fine-tuning

In the pre-training step, the AdamW[52] optimizer with a learning rate of 10^{-4} and weight decay of 10^{-2} was used in all three tasks. The model was trained with a batch size of 1,024 during 100 epochs. The pre-training dataset was randomly split into training, validation and test sets of size 800,000, 126,611 and 100,000, respectively. The learning rate was warmed up during the first 5% of the total epoch and was then linearly decayed to zero for the remaining epochs.

For fine-tuning, MOFTransformer is trained to predict the desired properties with the model initialized by the converged weights from the pre-trained model. By adding a single dense layer to the class token, all model weights are fine-tuned to predict desired properties of MOFs. Given that the relatively small datasets are used during the fine-tuning step, the model was trained with a batch size of 32 during 20 epochs whose optimizer and learning rates are the same as those of the pre-training step. The fine-tuning dataset was randomly split into training, validation and test sets in a ratio of 0.8:0.1:0.1. For scaling the target properties, a standardization method was adopted. Training details of the three baseline models for comparison of the fine-tuning models are explained in Supplementary Note 5.

Data availability

Data used in this work are available via Figshare (https://doi.org/10.6084/m9.figshare.21155506)[53]. This provides the pre-trained model and the atom-based graph embeddings and the energy-grid embeddings used as inputs of the MOFTransformer for CoREMOF, QMOF database as well as fine-tuning data. In addition, The UFF-optimized CIF files of hMOFs used in this work are available via Figshare (https://doi.org/10.6084/m9.figshare.21810147)[54].

Code availability

The MOFTransformer library is available at https://github.com/hspark1212/MOFTransformer ref. [55]. Documents for the library are available at https://hspark1212.github.io/MOFTransformer, which provides up-to-date documentation for pre-training, fine-tuning and feature importance analysis with MOFTransformer. For the sake of reproducibility, all results in this paper are obtained from version 1.0.1 of the MOFTransformer library, which is available at https://pypi.org/project/moftransformer/1.0.1.

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Author contributions

Y.K. and H.P. contributed equally to this work. Y.K. and H.P. developed MOFTransformer and wrote the paper with J.K. The paper was written through the contributions of all authors. All authors have given approval for the final version of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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